Specification of Business Processes

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1. Business Goals of the Organization

EcoVolt is a renewable energy generation facility utilizing wind and solar power. The main goal of the farm is to maximize energy production while minimizing downtime and operational costs. To achieve this goal, the management monitors two main metrics: daily energy generation output and the frequency of equipment maintenance incidents. Currently, the biggest problem for management is answering why energy production fluctuates significantly during similar weather conditions, and determining how to optimize maintenance schedules effectively.

Measurable Goals

- Increase overall energy production by 2% annually.
- Reduce turbine and solar panel downtime by 5% compared to the previous year.

2. Business Processes

Energy Production Monitoring

a. Description

The process of energy production monitoring involves continuously capturing power generation data from individual solar panels and wind turbines. Smart meters installed on each unit read the energy generated in kWh and transmit this data to the central monitoring system. The system logs the production amount, timestamp, and operational status of each unit.

b. Typical Questions

- Compare the energy produced by wind turbines versus solar panels this month.
- What are the most efficient turbines in this quarter?
- Calculate the average daily energy production for the last year.

- Compare the energy production of the current month with the production of the previous month.
- What is the total energy produced during extreme weather events?

c. Data

All production data is extracted from the primary monitoring system – PowerMaster. The system stores an asset hierarchy detailing the manufacturers and models of equipment, including the unit type (wind/solar) and maximum capacity. It maps individual physical units to geographical zones across the farm. Smart meters on each unit generate a continuous production log, recording the exact kWh produced at specific timestamps. In addition, historical weather data (temperature, wind speed, weather conditions) is provided hourly by a local meteorological institute and stored in a separate CSV file, which can be integrated with production logs via dates and zones.

Equipment Maintenance

a. Description

The equipment maintenance process handles routine and emergency repairs of power generation units. When a unit's efficiency drops or it breaks down, a technician is dispatched. The technician logs the issue type, parts replaced, time taken for the repair, and the downtime duration of the unit in the maintenance system.

b. Typical Questions

- Compare the number of maintenance interventions required for solar panels versus wind turbines.
- Which technicians completed the most repairs this month?
- Calculate the total downtime hours for each unit in the last year.
- Are there specific weather conditions that increase the frequency of emergency repairs?
- Compare the average repair time in the current month with the previous month.

c. Data

All maintenance data is fetched from the maintenance tracking system named “ServiceDesk**”**. Each intervention is registered as a maintenance ticket by a worker, which records the unit being serviced (using the same UnitID as in PowerMaster), the technician

assigned, the type of intervention (routine or emergency), the issue category (e.g., inverter failure, blade damage, panel soiling), the date and time the unit stopped working, the date and time it was restored, and the time the technician spent on the repair. Downtime is calculated as the difference between the restore time and the stop time. Spare parts used during the repair are recorded as separate ticket items (part name, quantity, cost). The system also stores a list of technicians with their specialization (wind or solar). Because tickets use the same UnitID, maintenance data can be integrated with PowerMaster production data and, through the unit's zone and the date, with the weather data.